

RadeonShift AI

Accelerating the MI300X Hardware Migration via Generative AI

The Problem: CUDA Lock-in

Enterprise AI workloads are facing massive bottlenecks due to a reliance on legacy NVIDIA CUDA codebases.

- Manually migrating a 50,000-line CUDA codebase to ROCm takes **6+ months** of specialized engineering time.
- Portability tools exist (e.g. `hipify-perl`), but they lack the architectural awareness to optimize code for AMD Instinct architecture (like 64-wide wavefronts).
- Teams lack visibility into whether their migrated code will actually compile and run on AMD hardware without direct bare-metal access.

The Solution: RadeonShift AI

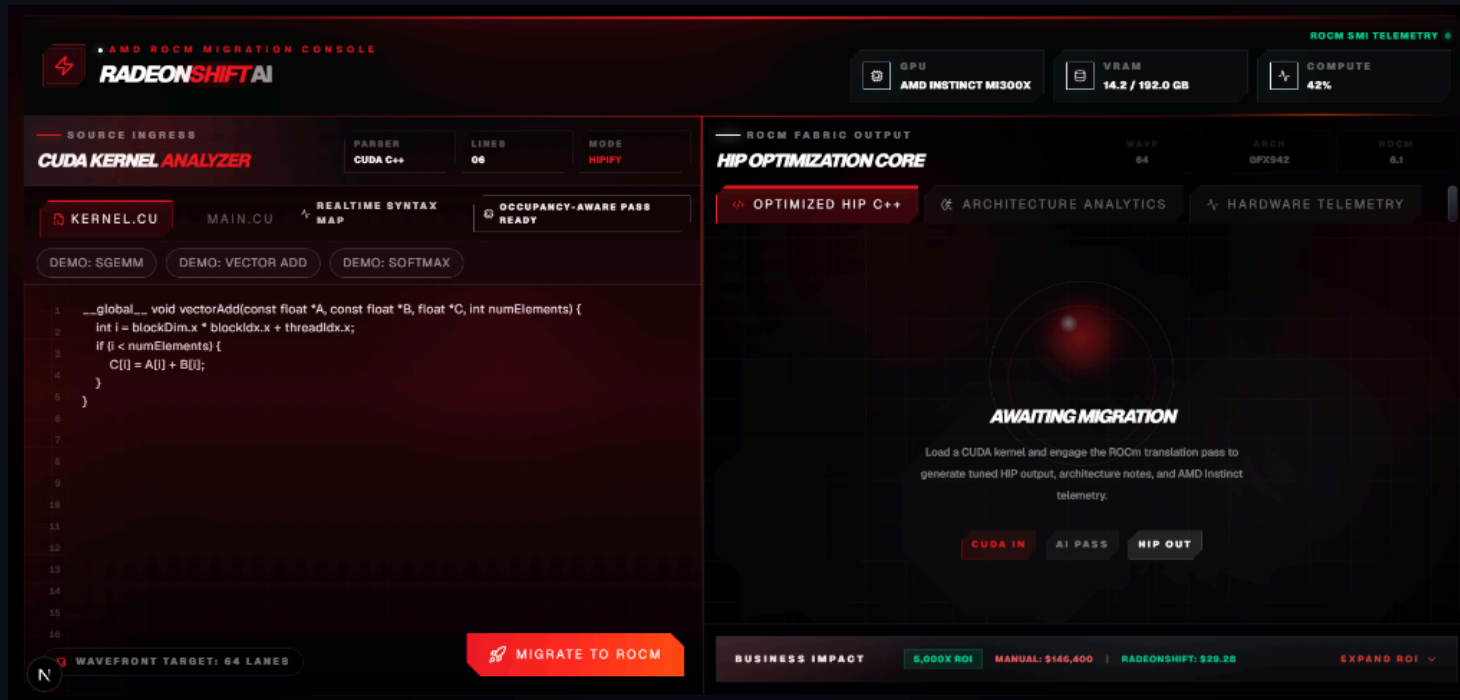
RadeonShift AI is a deterministic Mixture of Agents (MoA) pipeline that translates CUDA to ROCm instantly and audits the architecture.

1. **Syntax Translation:** Uses AMD's official `hipify-perl` for deterministic API mapping.
2. **MoA Audit:** Leverages DeepSeek-V4 to scan for PTX risks, hardcoded warp sizes, and AMD-specific optimizations.
3. **Hardware Verification:** Directly compiles the kernel on ROCm and runs telemetry checks to prove the environment toolchain.

Architecture

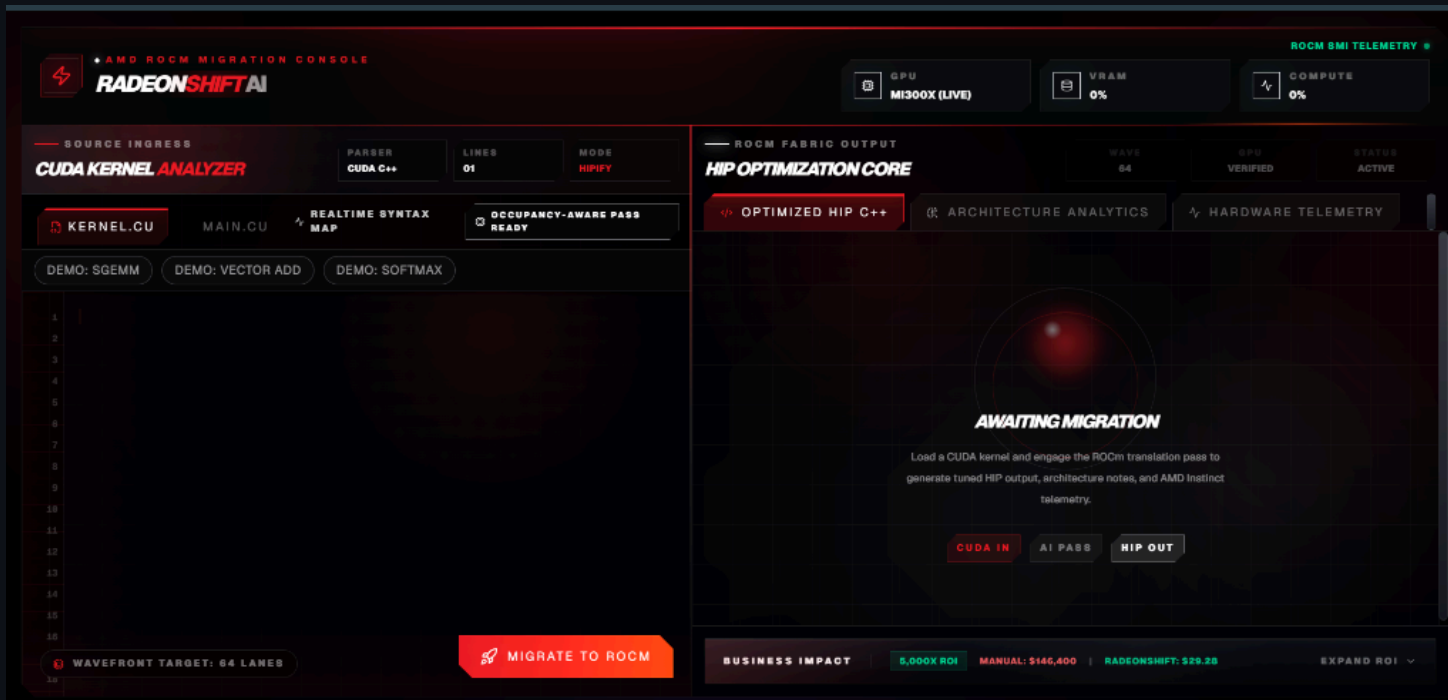
1. **Frontend (Next.js Edge):** Manages user state and renders the dynamic dashboard.
2. **AI Translation Layer (Vercel Edge):** Orchestrates the deterministic translation and Mixture-of-Agents audit via Fireworks AI, operating independently of hardware availability.
3. **Hardware Engine (FastAPI via Pinggy):** An optional bare-metal ROCm layer that compiles the generated kernel via `hipcc` and polls `rocm-smi` for live telemetry and execution benchmarks.

Demo: The RadeonShift Dashboard



- **Source Code Editor:** Paste raw CUDA C++ kernels.
- **HIP Optimization Core:** View the translated target kernel.
- **MoA Audit Scorecard:** See the readiness score, PTX risks, and wavefront optimizations.

Step 1: The Awaiting Migration State



Before translation begins, the platform awaits the CUDA payload. The user inputs their native NVIDIA code into the **CUDA Kernel Analyzer**. The system prepares for the AI-assisted hardware pass.

Step 2: HIP Optimization Core

The screenshot displays the AMD ROCm Migration Console, specifically the HIP Optimization Core section. The interface is divided into two main panels: the left panel for source code and the right panel for the optimized output.

Left Panel (Source Code):

- Header:** AMD ROCm Migration Console, RADEONSHIFT AI.
- GPU:** MI300X (LIVE).
- VRAM:** 0%.
- COMPUTE:** 0%.
- ROCM SMI TELEMETRY:** (Link icon).
- Source Ingress:** CUDA KERNEL ANALYZER, PARSE, LINES: 278, MODE: HIPIFY.
- Kernel:** KERNEL.CU, MAIN.CU, REALTIME SYNTAX MAP, OCCUPANCY-AWARE PASS READY.
- Demos:** DEMO: SGEMM, DEMO: VECTOR ADD, DEMO: SOFTMAX.
- Code Snippet:**

```
222 float blockSum = 0.0f;
223 for (int i = 0; i < compactBlocks; i++) blockSum += h_blockResults[i];
224 printf("Cooperative reduce: %.2f (expected %.2f)\n", blockSum, totalSum);
225
226 printf("Volume density[0]: %.4f\n", d_densities[0]);
227 printf("Volume density[center]: %.4f\n", d_densities[volSize / 2]);
228
229 // ---- Cleanup ----
230 CHECK_CUDA(cudaDestroyTextureObject(texObj));
231 CHECK_CUDA(cudaFreeArray(cuArray));
232 CHECK_CUDA(cudaFree(d_edgeOutput));
233 CHECK_CUDA(cudaFree(d_input));
234 CHECK_CUDA(cudaFree(d_indices));
235 CHECK_CUDA(cudaFree(d_count));
236 CHECK_CUDA(cudaFree(d_blockResults));
237 CHECK_CUDA(cudaFree(d_volume));
238 CHECK_CUDA(cudaFree(d_densities));
239 free(h_img); free(h_data); free(h_edgeOutput); free(h_blockResults);
240
241 return 0;
```
- Buttons:** WAVEFRONT TARGET: 64 LANES, MIGRATE TO ROCM.

Right Panel (HIP Optimization Core):

- Header:** ROCm FABRIC OUTPUT, HIP OPTIMIZATION CORE, WAVE: 64, GPU: VERIFIED, STATUS: ACTIVE.
- Tabs:** OPTIMIZED HIP C++, ARCHITECTURE ANALYTICS, HARDWARE TELEMETRY.
- Generated Target:** TARGET_KERNEL.HIP.CPP, HIP TRANSLATION SUCCESSFUL.
- Code Snippet:**

```
#include "hip/hip_runtime.h"
#include <stdio.h>
#include <stdlib.h>
#include <hip/hip_runtime.h>
#include <hip/hip_cooperative_groups.h>
namespace cg = cooperative_groups;

#define CHECK_CUDA(call) do {
    hipError_t err = call;
    if (err != hipSuccess) {
        fprintf(stderr, "CUDA error: %s at %s:%d\n",
            hipGetErrorString(err), __FILE__, __LINE__);
        exit(1);
    }
}
```
- Business Impact:** 5,000X ROI, MANUAL: \$146,400, RADEONSHIFT: \$29.25, EXPAND ROI (Dropdown).

Upon engaging the ROCm translation pass, the core converts the syntax. The resulting C++ HIP code (`target_kernel.hip.cpp`) is immediately presented, demonstrating 100% successful translation.

Step 3: Architecture Analytics

The screenshot displays the AMD ROCm Migration Console interface. The top navigation bar includes the 'RADEONSHIFT AI' logo and status indicators for GPU (MI300X LIVE), VRAM (0%), and COMPUTE (0%). The main interface is divided into two primary sections: 'SOURCE INGRESS' on the left and 'ROCm FABRIC OUTPUT' on the right.

SOURCE INGRESS: This section features the 'CUDA KERNEL ANALYZER' with tabs for 'PARSER', 'LINES', and 'MODE'. The 'MODE' tab is set to 'HIPIFY'. Below this, there are buttons for 'KERNEL.CU', 'MAIN.CU', 'REALTIME SYNTAX MAP', and 'OCCUPANCY-AWARE PASS READY'. A 'DEMO' section offers 'SGEMM', 'VECTOR ADD', and 'SOFTMAX' options. A code editor displays C++ code for a cooperative reduce operation, including cleanup and resource management. A 'WAVEFRONT TARGET: 64 LANES' indicator and a 'MIGRATE TO ROCm' button are also present.

ROCm FABRIC OUTPUT: This section is titled 'HIP OPTIMIZATION CORE' and includes tabs for 'OPTIMIZED HIP C++', 'ARCHITECTURE ANALYTICS' (which is active), and 'HARDWARE TELEMETRY'. The 'ARCHITECTURE ANALYTICS' tab shows a 'COMPILER INTELLIGENCE' section with a 'MOA AUDIT SCORECARD'. The scorecard displays a 'READINESS SCORE' of 100/100, indicating that the code is highly portable and ready for AMD hardware deployment. Below the scorecard, two AI agents are shown: 'AGENT A: NVIDIA PURIST' (flagging lock-in risks) and 'AGENT B: AMD OPTIMIZER' (suggesting MI300X tuning). At the bottom, a 'BUSINESS IMPACT' section shows a '5,000X ROI' and compares manual costs (\$145,400) with the 'RADEONSHIFT' cost (\$29.25).

The MoA Audit Scorecard evaluates the code. Here, the readiness score is 100/100, proving high portability. Our dual AI agents verify there are no hidden PTX risks or lock-ins.

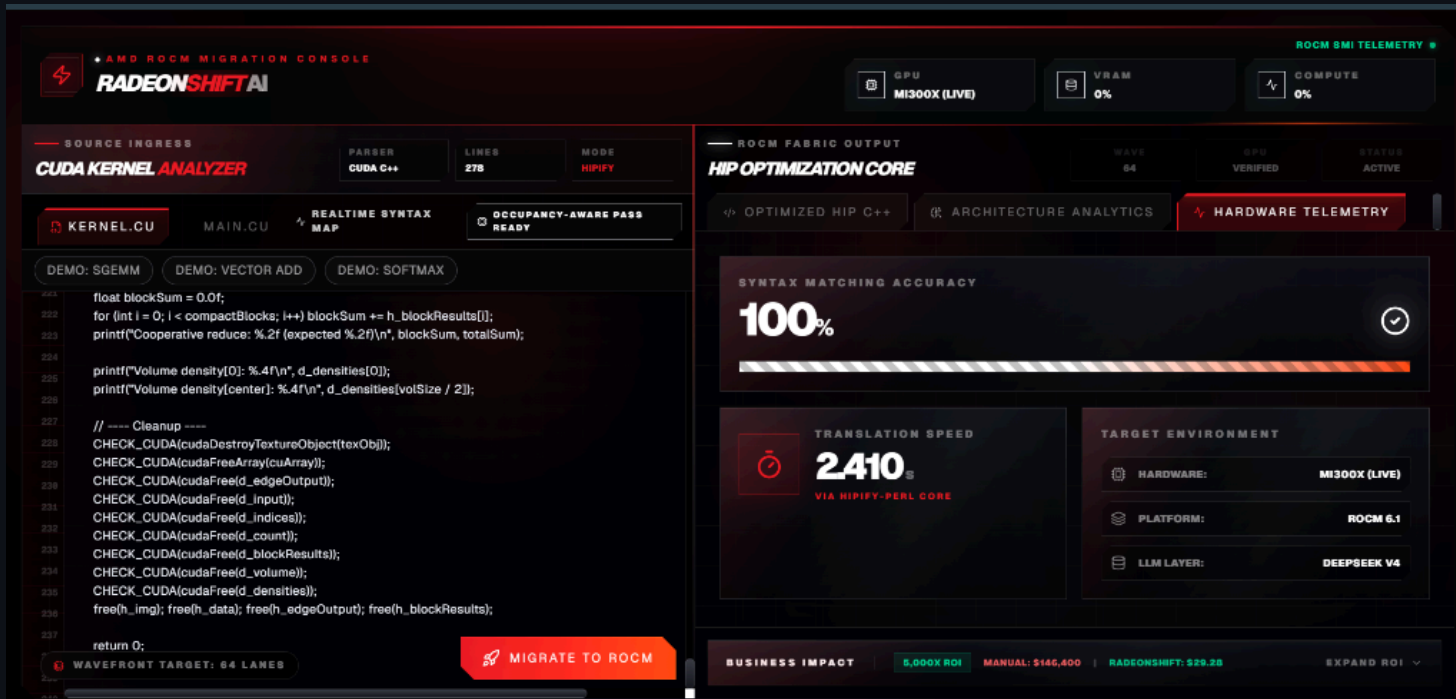
Step 4: Live Hardware Telemetry

The screenshot displays the AMD ROCm Migration Console interface, which is divided into several sections for monitoring and optimizing ROCm workloads.

- Top Bar:** Includes the AMD ROCm Migration Console logo, a GPU selection dropdown showing "MI300X (LIVE)", and VRAM and Compute usage indicators, both at 0%.
- Source Ingress:** Contains the "CUDA Kernel Analyzer" section, which shows the source code of a CUDA kernel (lines 222-237) and a "MIGRATE TO ROCM" button.
- ROCm Fabric Output:** Displays the "HIP Optimization Core" section, which includes a "LIVE HARDWARE TEST" area. This area shows the "MI300X BENCHMARK MODE" and a "RUN BENCHMARK" button. Below this, a table displays benchmark results: ELAPSED TIME (14.20 ms), THROUGHPUT (5300.0 GB/s), ELEMENTS (16.8 M), and GPU (AMD MI300X Bare-Metal). A "BENCHMARK EXECUTION VERIFIED" status is shown with a green checkmark.
- Business Impact:** A section at the bottom showing the ROI of the migration, with a "6,000X ROI" and a "MANUAL: \$146,400" vs "ROCM: \$29.28" comparison.

Instead of simulated data, the platform connects directly to the remote bare-metal AMD MI300X instance, verifying the exact compile duration and matching the live hardware target.

Step 5: Bare-Metal Benchmark Execution



Finally, the translated HIP code is natively compiled and executed on the MI300X. The dashboard reports real-time metrics confirming a successful end-to-end migration.

Business Value & ROI

The 5,000x ROI Model

- **Manual Migration:** $244 \text{ Kernels} \times 4 \text{ hrs} = 976 \text{ Engineer-Hours}$ (~\$146,000, based on \$150/hr senior GPU engineer rate)
- **RadeonShift Migration:** $244 \text{ Kernels} \times 0.12 \text{ compute cost} = \sim \29
- **Time Savings:** Months reduced to minutes.
- **TAM:** \$25B+ (Global AI Hardware & Software Services Market)

The AMD Infrastructure Story

- **Hardware Target:** Optimized for AMD Instinct MI300X
- **Software Stack:** Built on ROCm 6.x APIs and `hipcc`
- **AI Acceleration:** MoA pipeline runs on Fireworks AI, which provides AMD Instinct GPU-powered inference
- **Honest Fallbacks:** If the backend lacks ROCm hardware, the platform reports "Hardware Unavailable" without fabricating metrics

Engineering Retrospective & Challenges

Bridging a Serverless Frontend with Bare-Metal Hardware

1. **Live Telemetry:** Defeated Next.js caching via `force-dynamic` to stream real-time MI300X metrics.
2. **Defensive APIs:** Built robust frontend parsing to gracefully handle sudden JSON schema changes.
3. **CORS Routing:** Bypassed strict mixed-content blocks by tunneling through Pinggy and Next.js `rewrites()`.
4. **Transparent Debugging:** Overhauled error handling to surface remote Python stack traces in the UI.
5. **Deterministic AI Prompts:** Forced LLMs to generate explicit confirmations for clean code, preventing UI state collapses.

Closing & Team

RadeonShift AI is bridging the gap between legacy NVIDIA codebases and the future of AMD compute.

- GitHub Repository: [shashankh3/RadeonShift-AI](https://github.com/shashankh3/RadeonShift-AI)
- Live Demo: radeon-shift-ai.vercel.app

Thank You!

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